NB SG 5 – Sinonasal and Parapharyngeal Cases

Required materials and equipment:

1. A copy of the clinical cases
2. A copy of the research article
3. An electronic copy or hard copy of your Netter's Atlas (required) and lecture notes (optional)

How to prepare and what to expect from this session

1. Research Article: 15 minutes

Use library resources or open resources (e.g. Google Scholar) to find a peer reviewed article on Horner's syndrome. Use the following objectives as a guide to review the selected paper. Once you are able to answer most of these questions, then you have successfully completed your review and are fully prepared for discussion.

a) How did you find the article?
b) Why did you choose the specific article?
c) What are the typical complaints associated with the syndrome?
d) With these symptoms what other possible diagnosis should be explored?
e) What questions did you have while reading the paper?

2. Clinical Case Discussion: 80 minutes

a) Use your lectures, DLAs, textbooks and Sketchfab models to answer the clinical case questions listed below. These questions will be discussed with your peers and clinical facilitators during the small group. At the end of the session, all students are responsible for having completed all the learning objectives listed for that session. If a group is not able to finish all learning objectives in the allotted time, students have the individual responsibility for learning them on their own. If a group finishes all learning objectives before the allotted time has ended, the students must reinforce the learned concepts by inviting interactions with Facilitators, peers and subject experts.

b) How to use SketchFab:
I) Open the links below

> a. IT PT Fossae: https://sketchfab.com/3d-models/nerve-content-of-the-it-ptfossae-6cbe72f8e0fe4e9a9321fc6f81b2e46d
> b. IT PT Fossae Box Model: https://sketchfab.com/3d-models/content-andboundaries-of-the-it-pt-fossae-2b139ac74a7f4f2a9c218c56029848be
> c. Spaces of the Head and Neck: https://sketchfab.com/3d-models/spaces-ofthe-head-and-neck-anatomical-modelace11450d00d4653b4c34bf53bfc70a4

II) Use the mouse or trackpad to rotate and zoom in/out to view the 3D models.
III) Utilize the 3D visualization with the cases in the small group session to better understand the anatomical structures.

Clinical Case Questions:

Case A

Objectives

SOM.MK.I.BPM2.1.NB.1.ANAT.0101	Describe the blood supply and lymphatic drainage of the nasal cavity
SOM.MK.I.BPM2.1.NB.1.ANAT.0102	Describe the paranasal sinuses, the bones in which they are found and where each opens into the nasal cavity
SOM.MK.I.BPM2.1.NB.1.ANAT.0103	Describe the sensory and parasympathetic innervation of the nasal cavity
SOM.MK.I.BPM2.1.NB.1.ANAT.0108	Explain the clinical significance of Kiesselbach's area in relation to epistaxis

A 27-year-old man comes to the physician because of headache, fever and blocked nose of 7 days duration. He also complains of pain in his upper teeth and swollen neck glands. He explains that he had just returned home after competing in a surfing competition in Hawaii during which he was rolled onto the shore and sand by a large wave. Physical examination shows swollen nasal mucosa with green mucous production and pain on palpation inferior to the orbit. A diagnosis of sinusitis is made.

1. What is the sensory innervation to the nasal cavity?

2. What is the likely reason for the tooth pain?

3. Which ganglion contains the cell bodies of the postganglionic parasympathetic fibers responsible for the increased nasal secretions?

One week later the patient sneezes and expels a large amount of sand from his nose. In an attempt to clean out all the sand his nose starts to bleed.

4. Where was the sand most likely lodged?

5. Explain the drainage of the paranasal sinuses into the nasal cavity?

6. Describe the blood supply of the nasal septum.

7. What is Kiesselbach's area and why is it clinically important?

Case B

Objectives

SOM.MK.I.BPM2.1.NB.1.ANAT.0129	Explain the sensory and motor innervation of the nasopharynx, oropharynx, and laryngopharynx
SOM.MK.I.BPM2.1.NB.1.ANAT.0127	List the contributions to the pharyngeal nerve plexus
SOM.MK.I.BPM2.1.NB.1.ANAT.0126	List the general attachments, innervation, and functions of the muscles of the pharynx
SOM.MK.I.BPM2.1.NB.1.ANAT.0138	Describe the sensory and mucous innervation of the larynx
SOM.MK.I.BPM2.1.NB.1.ANAT.0137	List the attachments, innervations, and functions of the muscles of the larynx

At a family dinner, a 35-year-old father who was eating fish gasped and said that he felt that a bone was stuck in his throat. He was taken to the hospital where the bone was removed from the piriform fossa. Give explanations for the following:

1. Is the piriform fossa in the pharynx or in the larynx?
2. In removing the bone, what nerve in the mucosa of the piriform fossa must the physician avoid injuring?
3. What type of nerve fibers course in this nerve, and where is the distribution?
4. What could be the consequences if this nerve were injured during removal of the bone?
5. What is the innervation of muscles of the pharynx?
6. Which nerve(s) supply the muscles of the larynx?
7. What is the sensory innervation of the laryngeal mucosa above and below the vocal folds?

Case C

Objectives

SOM.MK.II.BPM2.1.NB.5023	Differentiate the functional components and pathways of the glossopharyngeal and vagus nerves based on their extracranial courses and fiber types.
SOM.MK.II.BPM2.1.NB.5024	Compare and contrast the nuclei and ganglia associated with the glossopharyngeal and vagus nerves, relating their anatomical positions to their functional roles.
SOM.MK.II.BPM2.1.NB.5025	Analyze how the distribution and branching patterns of the glossopharyngeal and vagus nerves contribute to their respective clinical presentations when damaged.
SOM.MK.II.BPM2.1.NB.5026	Appraise the role of each nerve's ganglia in sensory transmission and autonomic regulation, particularly in relation to reflex arcs.

A 58-year-old man comes to the emergency department because of 2 weeks of progressive hoarseness and difficulty swallowing. His medical history is significant for poorly controlled hypertension and smoking 1 pack of cigarettes daily for 30 years. On examination, he has weakness in turning his head to the right and raising his left shoulder. Examination of the oral cavity shows absent gag reflex on the left, deviation of the tongue toward the left on protrusion, and decreased sensation over the posterior third of the tongue. MRI of the neck shows an infiltrative mass in the parapharyngeal space extending posteriorly toward the carotid sheath.

1. Explain how the MRI findings correlate with the clinical presentation in this patient.

2. List the cranial nerves most likely affected by this lesion. For each, describe major extracranial branches, functional components, and expected clinical signs if functionality is impaired.

3. What are the anatomical relationships between the jugular foramen, carotid sheath, and parapharyngeal space? Why is this region vulnerable to compressive lesions?

4. Which cranial nerve lesion would explain the loss of sensation in the posterior third of the tongue? Include discussion of relevant nuclei and ganglia.

5. The patient has hoarseness and difficulty swallowing. Which specific cranial nerve branches are involved, and what are their motor targets?

6. Which nucleus is primarily responsible for parasympathetic outflow in the vagus nerve, and what is its role in autonomic regulation? How could this lesion potentially alter vagal reflexes?

7. Describe how the glossopharyngeal and vagus ganglia contribute to visceral sensory transmission. Relate these ganglia to the clinical presentation seen in this patient.

8. The patient shows tongue deviation on protrusion. Which cranial nerve is responsible for this finding, and how is it anatomically related to the jugular foramen region?

9. The patient has weakness in turning his head to the right and difficulty elevating the left shoulder. Which nerve is responsible, and what are its origins and course through the neck?